

Exercises for Chapter 0

0.1.1

Given $v \in [0^\circ, 90^\circ]$.

- a) Show that $\sin v = \sin(180^\circ - v)$.
- b) Show that $\cos v = -\cos(180^\circ - v)$

0.1.2

Find the area of $\triangle ABC$ when

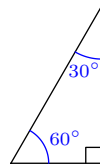
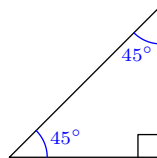
- a) $\angle A = 60^\circ$, $AB = 5$ and $AC = 7$.
- b) $\angle B = 18^\circ$, $AB = 4$ and $BC = 3$. $\left(\sin 18^\circ = \frac{\sqrt{5}-1}{4}\right)$
- c) $\angle A = 75^\circ$, $\angle B = 60^\circ$, $AC = \sqrt{6}$ and $BC = \sqrt{3} + 1$

0.1.3

- a) Prove the area theorem.
- b) Prove the sine theorem.

0.1.4

- a) Show that $\cos 45^\circ = \frac{\sqrt{2}}{2}$.
- b) Show that $\sin 30^\circ = \frac{1}{2}$.
- c) Show that $\cos 30^\circ = \frac{\sqrt{3}}{2}$.

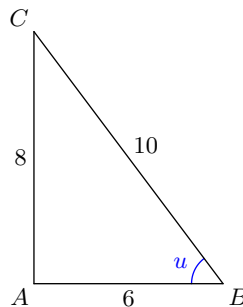


0.1.5 (1TV23D1)

A right triangle has sides 6, 8, and 10. See the figure to the right.

Show that

$$(\sin u)^2 + (\cos u)^2 = 1$$

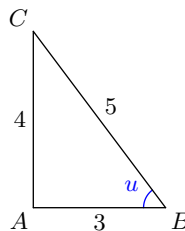


0.1.6 (1TH22D1)

Given the triangle to the right.

Show that

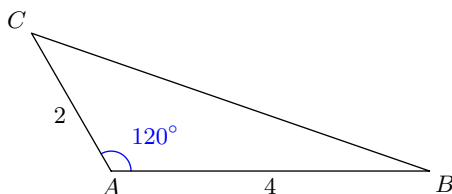
$$\frac{\sin u}{\cos u} = \tan u$$



0.1.7

Show that $\tan v = \frac{\sin v}{\cos v}$.

0.2.1 (1TH21D1)



Given the triangle above. Determine the length of side BC .

0.2.2

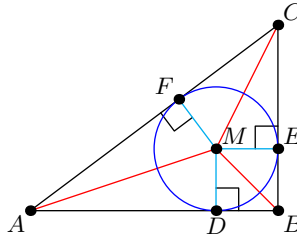
Given a triangle with sides a , b , and c and an inscribed circle with radius r . Explain why the area of the triangle is given as

$$\frac{1}{2}(a + b + c)r$$

0.2.3

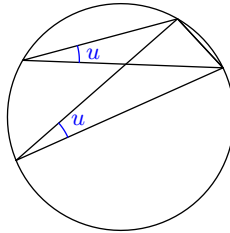
Let $a = BC$, $b = AC$, $c = AB$ and $DM = r$.

- Show that $r = \frac{ac}{a+b+c}$.
- Show that $2r = a + c - b$.
- Use the expressions from tasks a) and b) to find b^2 expressed by a and c . What is this formula called?



0.2.4

Explain why, from Rule ??, it follows that two angles spanning the same arc are equal in size.



0.2.5

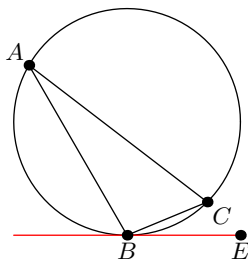
- Show that Thales' theorem¹ follows from Rule ??.
- Given a right-angled triangle $\triangle ABC$ with hypotenuse AB . Which of $\Rightarrow$, $\Leftarrow$ and $\Leftrightarrow$ should replace ??? below to describe the *inverse* case of Thales' theorem.

$C = 90^\circ$??? AB is a diameter in the circumscribed circle of $\triangle ABC$

¹See MB.

0.2.6

The red line is tangent to the circle. Show that $\angle BAC = \angle EBC$.



Ponder 1

(1TH21D1)

A triangle has a perimeter of 12, and one side of the triangle is 2.
Determine the area of the triangle.

Ponder 2

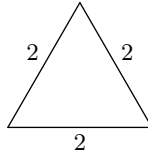
(1TV21D1)

Sort the values in ascending order.

$$\sin 60^\circ \qquad \left(\frac{3}{4}\right)^{-1} \qquad \sin 160^\circ \qquad \lg 1$$

Ponder 3

An equilateral triangle has sides of length 2.



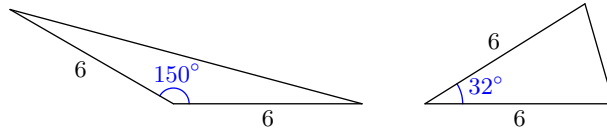
Use the triangle to show that

$$\cos 60^\circ = \frac{1}{2}$$

Ponder 4

Which of the two triangles has the larger area?

Remember to argue why your answer is correct.



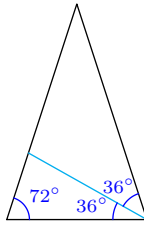
Ponder 5

Show that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Ponder 6

Show that $\sin 18^\circ = \frac{1}{4}(\sqrt{5} - 1)$. (Hint: See figure.)



Ponder 7

Prove the cosine theorem.

Ponder 8

Show that

$$\cos(u + v) = \cos u \cos v - \sin u \sin v$$

It is sufficient to examine the case where $v, u \in [0^\circ, 90^\circ]$.

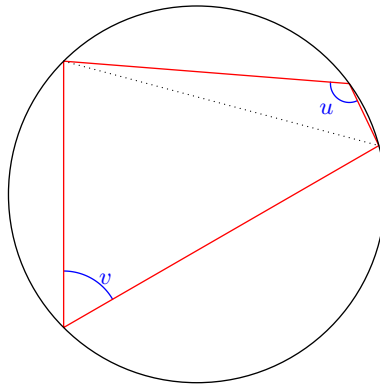
Ponder 9

Prove the converse case of Thales theorem (see [Exercise 0.2.5](#)).

Ponder 10

Show that

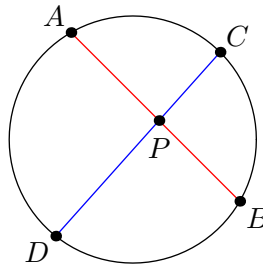
$$u = 180^\circ - v$$



Ponder 11

Show that

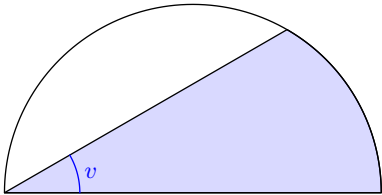
$$AP \cdot PB = DP \cdot PC$$



Note: This result is often called **the chord theorem**.

Ponder 12

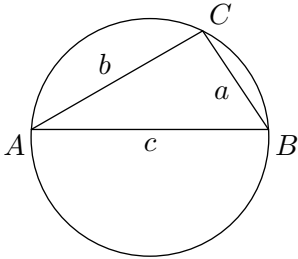
Let r be the radius of the semicircle. Express the area of the blue area in terms of v and r .



Ponder 13

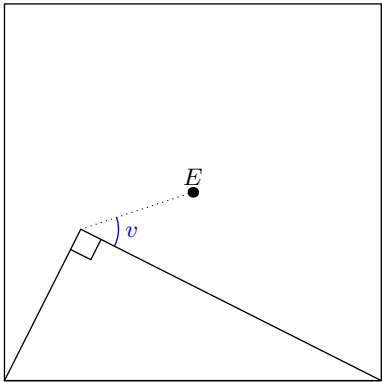
Let r be the radius of the circumscribed circle to $\triangle ABC$. Show that

$$r = \frac{abc}{4A_{\triangle ABC}}$$



Ponder 14

E is the midpoint of the square. Find the value of v .



Ponder 15

Show that

$$AB^2 = BC \cdot CD$$

